“Testing and Treatment Options Review for Robert Ellis” (Robert Ellis) [#39]

Brad Power
January 11, 2023

“My main goal is to find some way of identifying more targetable markers that might give me more treatment options.” Robert Ellis

“What other things should I be talking to my doctor about or see if I can get lined up for after the current treatment that I'm on, after I stop responding to my current treatment?” Robert Ellis

Meeting Summary

Robert Ellis is an advanced prostate cancer patient who is looking for guidance on his treatment options and strategy after he stops responding to his current treatment. In particular, he is interested in tests that could uncover biomarkers that would lead to targetable treatment options he could discuss with his doctor.

What is Robert’s current situation?

Robert was diagnosed in October 2017 with advanced prostate cancer – it had metastasized to his bones. Over the last five or six years he has been through many rounds of treatment: several androgen deprivation therapy (ADT) drugs, focused radiation, immunotherapy, a PARP inhibitor, and chemotherapy. His disease, as measured by Prostate Specific Antigen (PSA) tests, which he has gotten as frequently as every three weeks, has risen and fallen three times.

He is currently on a hormone therapy (Orgovyx), a PARP inhibitor (Rubraca), a urination drug (dutasteride), and supplements, including metformin. His PSA is currently low (1.7).

He is considering two treatment options for his next line of treatment: (1) radiation particles targeted at Prostate Specific Membrane Antigen (PSMA), called “Pluvicto” or “Lutetium 177”, and (2) bipolar androgen therapy, where testosterone is introduced to the cancer cells which have been bred to survive in a hormone-deprived environment.

Robert’s priority for treatment is quality of life – enjoying his life, choosing approved drugs, not riskier treatments, low side effects, and not choosing treatments that would take a lot of time or travel to access. He is interested in drug combinations, intermittent treatments, and rechallenging.

What additional testing options were suggested?

- Liquid biopsy, e.g., the Tempus xF+ test, can provide biomarkers for targeted treatments.
- Proteomic analysis, e.g., mProbe tissue analysis, can provide new biomarkers for targeted treatments.

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- Standard Uptake Values from a PSMA PET scan will indicate whether he is likely to have a response to PSMA-targeted radiation therapy.
- Current status of PDL1/PD1, tumor mutation burden, and MSS/MSI, can help predict whether he will likely be responsive to immunotherapy.
- Whole genome and RNA sequencing can identify biomarkers which may be targetable.
- Bone biopsy, possibly the bone aspirate liquid biopsy being researched by Peter Kuhn, to be able to analyze bone tumors.

What additional treatment options were suggested?

- An Androgen Receptor (AR) degrader (ARV-110 or ARV-766 clinical trial), given the high expression of AR as noted in the Caris report.
- CAR-T, if there are indications that an immunotherapy might work.
- Drug combinations of abiraterone + olaparib + sulindac, carboplatin + degarelix + sulindac, apalutamide + lutetium lu 177 vipivotide tetraxetan + sulindac, abiraterone + olaparib, carboplatin + degarelix, and apalutamide + lutetium lu 177 vipivotide tetraxetan

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Meeting Notes

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SUMMARY KEYWORDS

psa, treatment, tissue, started, report, liquid biopsy, clinical trial, adt, robert, chemo, biopsy, test, cancer, respond, little bit, therapy, psm, scan, chemotherapy, metastases

SPEAKERS

Robert Ellis (62%), Brian McCloskey (10%), Rick Stanton (9%), Amit Gattani (6%), Brad Power (6%), Ally Perlina (4%), Mike Yancey (3%)

Brad Power

Today is the type of Prostate Cancer Lab meeting where we have one of the patients in our community review his case. Today we are going to hear from Robert Ellis.

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Robert is going to review his case and look for some feedback on ideas to help him with his testing and treatment decisions.

Robert Ellis (1:27)

I want to thank everybody for this opportunity to get your input and review my case. I apologize in advance that I don't have as much information as I'd hoped to have. I was scheduled to have a PSMA scan, and I had a biopsy in October that did not reveal any usable tissue.

My primary aims for this call are: I would love some guidance on specifically what kind of testing options I should talk to my doctor about to be able to identify some markers that might be targetable for treatment.

And the second thing is: right now in my conversations with my oncologist, I really have two treatment options in the wings. I'd like to know if there are some other treatment options that are

aligned with my sort of treatment principles which I'll share. And possibly combination treatments and things like that.

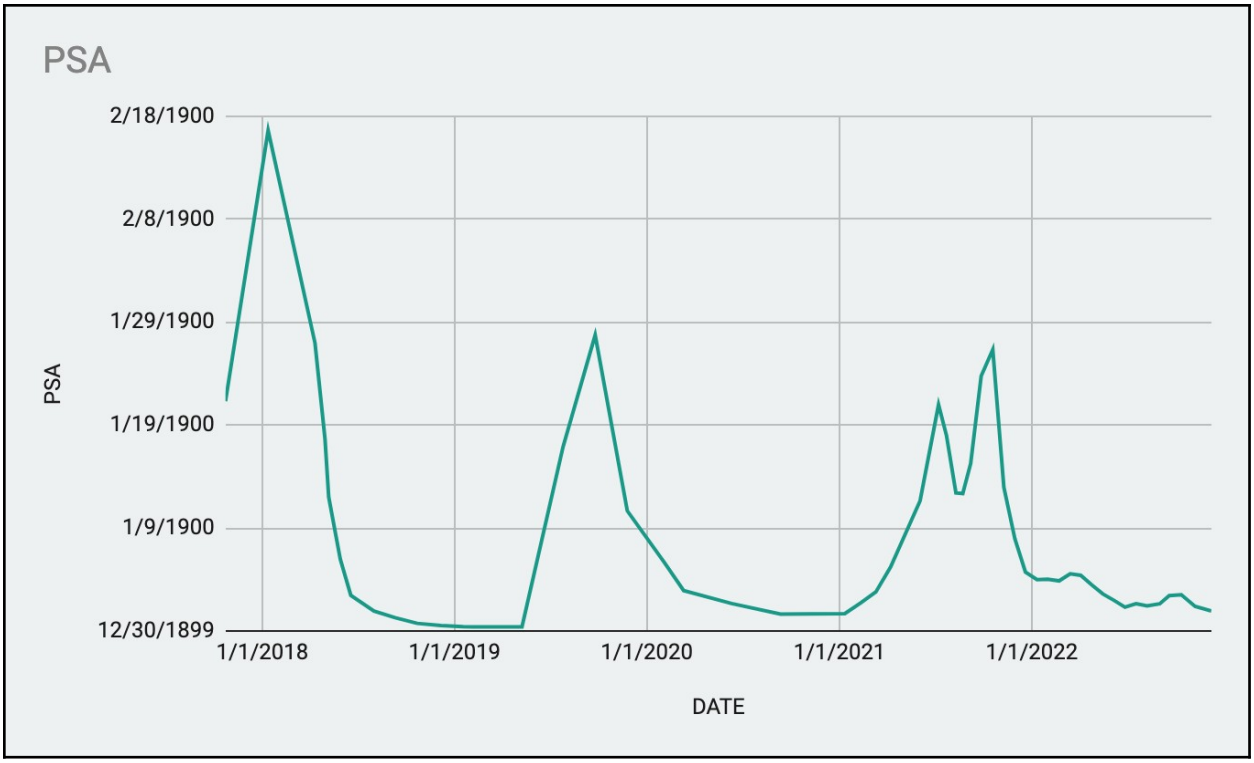
In other words, what other things should I be talking to my doctor about or see if I can get lined up for after the current treatment that I'm on, after I stop responding to my current treatment.

A little bit of a disclaimer: last night I saw Rick's email and looked at his depiction of his cancer history and was a little intimidated. And then I thought, I better go through all my notes on the Google Drive. And I found Brian's files, and that was pretty intimidating too. I'm going to show you an unfinished symphony. I'm inspired by the work that Rick and Brian have done, and I'm going to be more diligent about maintaining my records going forward.

Rick Stanton (3:40)

Don't forget Eric. He did a really cool depiction as well.

Robert Ellis (4:09)



Here is my spreadsheet with a complete record of my PSA. I have not added a legend like in Rick's beautiful charts.

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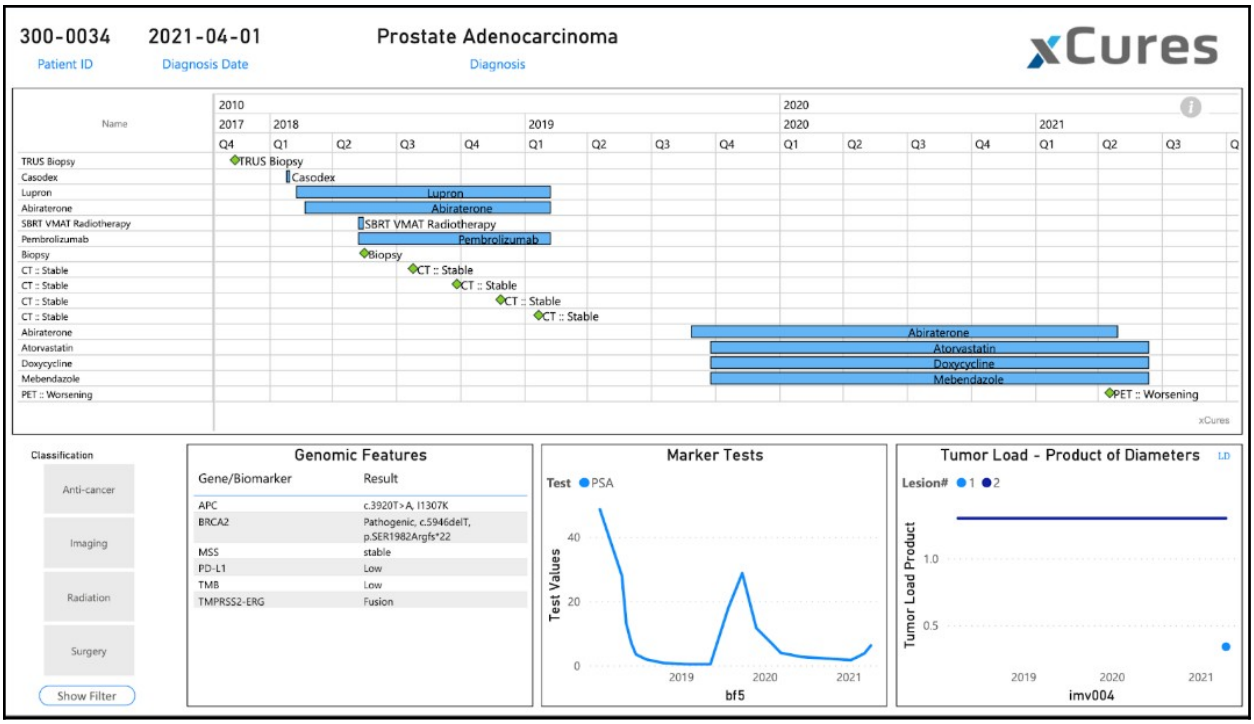

This chart here was provided by xCures. It's a little out of date. They put this together for me back in 2021. The reason I haven't had an updated one is I keep waiting to get new test results, but then I haven't gotten any.

I was diagnosed on October 23, 2017. When I had my first PSA, it was 22.3. The initial spike is because it took me a while before I began treatment. Shortly after that I went on a clinical trial. I started on ADT, which on the xCures chart was casodex followed by Lupron. Then abiraterone was added. I was on a clinical trial at UCSF, which was a combination of five treatments of focal radiation to the prostate, and 13 rounds of pembrolizumab every three weeks. I responded well. It was difficult to say whether that was the ADT or the clinical trial, but I responded pretty well. By May of 2019, my PSA was fractional at 0.384.

When I was at the end of this clinical trial, they stopped everything. I had no treatment at all. I felt really good for about three or four months, but at around five months I started to have pain again. I have metastases to my tissue of tuberosities (a large prominence on a bone usually serving for the attachment of muscles or ligaments). When I had the next PSA, I was at 28.73. So they started me on ADT again, Lupron and abiraterone, and then responded to that. At the beginning of 2021 I started to become castrate resistant. I was on the same treatment but my PSA was rising.

The next drop is where I started dutasteride, which is known to give a false PSA drop. It showed my PSA dropped when I began dutasteride, but it was not. It wasn't really treating the cancer. It was a false reading, and it spiked again after that.

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I began chemotherapy in October of 2021. The chemotherapy was a combination of carboplatin, because I have BRCA2, and docetaxel, initially, at a total of 16 rounds. Three of those were carboplatin and cabazitaxel. I asked to try cabazitaxel to see if the side effects were any less onerous than the docetaxel. They were worse: more neuropathy in my feet, which I still have. But I responded pretty well to chemotherapy for some time.

When my PSA started to go up again, we stopped the chemotherapy, and I began Rubraca (rucaparib, a PARP inhibitor). That was partly because earlier xCures had recommended a clinical trial that was a combination of olaparib and radium 233. Olaparib had always been in the wings, but I was advised not to do a PARP inhibitor earlier because there was a new generation in the works, and Rubraca was one of those. By the time the chemotherapy began to fail, I started the Rubraca.

Just after I started the Rubraca I managed to get my oncologist Dr. Koontz to speak with Dr. McKay about bipolar androgen therapy. It took a while for them to connect. In that interim, we started the Rubraca so that I would be on some treatment. I'm responding to it. So we'll continue until I stop responding. Dr. Koontz was open to considering bipolar androgen therapy. After that, though, I learned that there are some risks involved and the responses are mixed.

I have had CT scans during 2018 and 2019 Those were all stable. I was supposed to have a PSMA scan last fall, but for some reason they ran the wrong test. They did a PET CT instead of PSMA. I was supposed to have a PSMA this week on Monday, but the weather prevented that. So I'm having that on the 26th.

Brad Power (10:37)

In early 2022 you were on a PARP inhibitor. You were probably still on Lupron. What was the combination you were on? We often talked about drug combinations, but you were on a list of chemo and a PARP inhibitor and maybe ADT?

Robert Ellis (10:58)

I was not on the PARP inhibitor Rubraca. I started that after the chemo failed. All of this time I was on – I can't remember exactly when I started – Orgovyx (relugolix). I started using Orgovyx instead of Lupron. I stopped the abiraterone when I became castrate resistant. This year is when I started the chemo. In October of 2021 I was still on Orgovyx. This was all chemo and Orgovyx. There was a little bump.

By the way, I have no way of knowing, but during this time I was also experimenting with taking various supplements that were potential chemo sensitizers. I have no idea if they made any difference because I started to have a little rise in my PSA here. And then I got a little more mileage out of it. But then I just started failing – my PSA was rising.

So I stopped the chemo. I'm still on Orgovyx. I started Rubraca in November and October. By the way, I was also on an off label protocol from Care Oncology in the UK, which was a combination of atorvastatin, doxycycline, mebendazole, and metformin. I've cycled off and on the atorvastatin, doxycycline, and mebendazole, because sometimes I feel like I'm having some memory issues, and it's controversial whether or not atorvastatin or mebendazole might contribute to that. I still take Metformin. I have no idea whether or not any of that helps, but it doesn't seem to hurt.

Amit Gattani (13:14)

Where are you being treated? What is your plan B treatment facility or oncology team?

Robert Ellis (13:24)

I'm currently being treated at Pacific Cancer Care in Monterey. I live in Carmel Valley. I moved here in the summer of 2020 from San Francisco. I had my original biopsy at Sutter Health. I saw someone at Stanford before I had a scan and realized that I had metastases. I was talking to someone about a possible prostatectomy. And then we got the information that it was too late for that. Then I was in treatment at UCSF in a clinical trial with Dr. Lawrence Fong. I was treated there from 2018 to 2020. And now we're here at Pacific Cancer Care with Dr. Michael Koontz. I was talking to someone at Stanford regarding the clinical trial for Lutetium., but the timing was off. Lutetium had just been approved. So they couldn't put me in a clinical trial, and I couldn't get on it because they didn't have the facility yet to be able to treat patients outside of the clinical trial. And that's when I went on chemotherapy.

Mike Yancey (14:55)

You mentioned that you were on finasteride (a generic agent that is used to treat lower urinary tract symptoms, complications from these symptoms, and prostate cancer.). When did you stop, and how long were you on that?

Robert Ellis (15:00)

I'm still on dutasteride (a medication primarily used to treat the symptoms of benign prostatic hyperplasia, an enlarged prostate not associated with cancer). That's for urination, for polyuria (a condition where the body urinates more than usual and passes excessive or abnormally large amounts of urine each time you urinate).

I did not try Xtandi (enzalutamide) after Zytiga (abiraterone). There was some controversy. I'm fortunate in that I'm an executive coach. One of my clients is in the cancer space. And they were gracious enough to make an introduction for me to some doctors at MD Anderson and Harvard, and I actually got some free consultations. There were a couple of different recommendations, but the overall consensus was that chemo would be the most sensible next step. So I ended up doing chemo. One opinion was to try some different things, some other

ADT to see if I could get any more mileage out of ADT. I did try at one point. In early 2021, after the Lupron was failing, I did a round of firmer gun, which was the most painful experience. I don't know if any of you have been on Firmagon (degarelix, a hormonal therapy). The first round is two shots in your belly. I couldn't bend over for 10 days. It was extremely painful. I didn't respond to it. So that was pretty much the end of ADT, and then Orgovyx became available, and I got on that.

Brian McCloskey (16:48)

With Firmagon, a lot of it is in the art of how it's administered. The nurse really has to know how to put the needle in. I didn't have quite the reaction that you did, but I had significant bruising. It wasn't it wasn't fun, for sure.

Robert Ellis (17:06)

I don't recommend Firmagon.

Amit Gattani (17:18)

Where are your metastases?

Robert Ellis (17:24)

PET Scan Report

PET/CT 9/29/22				
1. Persistent increased (SUV 5.2) metabolic activity in the prostate.				
2. Only warm metabolic activity in the largest of the bony lesions (left acetabulum and right ischium). The other bony lesions show no activity.				
3. No evidence for increased metabolic activity in the previously seen left external iliac chain lymph nodes.				
4. This may be secondary to decreased sensitivity of F-18 FDG compared with PyL.				

This is my PET CT scan report from September of 2022. (This was supposed to be a PSMA test.)
It shows persistent increased metabolic activity in the prostate, and metabolic activity in the largest of the bony lesions in the left acetabulum (the socket of the hipbone, into which the head of the femur fits) and right ischium (the curved bone forming the base of each half of the pelvis). There is no evidence of increased metabolic activity previously seen in the left external iliac

chain lymph nodes. (The common iliac lymph nodes, four to six in number, are grouped behind and on the sides of the common iliac artery, one or two being placed below the bifurcation of the aorta, in front of the fifth lumbar vertebra.) On an earlier scan, there was just a little bit of activity in the lymph nodes. As far as I know, of course, it may be because this test wasn't as sensitive as the previous one. As far as I know, there's been no progression there. But that's one of the reasons I'm anxious to have a PSMA test.

I have had a PSMA test, and so I know that I have PSMA, and I am a candidate for Lutetium. But that's a more sensitive test, and I will have a better idea of whether or not there's been any progression to the disease.

It says that there is activity in the prostate, but I subsequently had a biopsy on October 18. They took six cores, three right and three left, and found no evidence of disease, which is kind of a mixed blessing. On the one hand, it may be that I have less of a burden there, but probably not. And unfortunately, it meant that I didn't get any tissue for testing.

Caris Report

Genomic Features 11/10/2017	
Gene/Biomarker	**Result**
APC	c.3920T>A, I1307K
BRCA2	Pathogenic, c.5946delT, p.SER1982Argfs*22
MSS	Stable
PD-L1	Low (negative)
TMB	Low
TMPRSS2-ERG	Fusion

A little history about the tissue. This is my Caris report from the original old tissue that was collected November 10, 2017. I have a BRCA2 mutation. I don't know what these other biomarkers mean. I copied them here if anybody knows whether or not any of this is useful,

I did a blood draw for the Tempus test where they take the liquid biopsy and compare it to the original tissue. What they found was that my original tissue was not usable, which is another reason why I wanted to get a fresh biopsy, but the biopsy in October was not usable. I'm not

exactly sure what happened to the liquid biopsy. At the time Dr. Koontz said something about it wasn't usable because they couldn't compare it to anything. I don't really understand that.

One of my questions is: where do I go from here as far as testing? I haven't had any usable tissue to test. With the PSMA scan, when I get it, it will be more sensitive and can tell me if there's more progression, or if I have more than metastases in the abutments or the lymph nodes. As of right now, it seems like my disease has been fairly stable over the past five years.

My Quality of Life Principles

• Manage risk (e.g., prefer approved drugs vs. early stage clinical trials)
• Low side-effect burden
• Least disruptive (e.g., travel, hospitalizations)
• Avoid becoming a full-time cancer patient

My priority for treatment is quality of life. I have a very full life right now, both work and personal. Kind of ironically, despite the cancer, I've been happier in the past five years than I've ever been. I'm really loving my life right now. I have a very full life. The criteria that I think of when I consider treatments are, first of all, I want to manage the risk. I prefer approved drugs versus early stage clinical trials, which might involve more risk. For example, when I was in conversation with Stanford about the lutetium clinical trial, they told me about another clinical trial. I forget what it was, but it was stage one, so there was some risk involved. It would have required hospitalization and so forth. I'm looking for the least disruptive treatments at this point, with a low side effect burden. Chemotherapy was the most onerous treatment that I've been under. It was certainly tolerable. But if I think about more radical treatments, how disruptive side effects would be is definitely a consideration. There are a lot of clinical trials that I have been made aware of but require travel and things like that. I would avoid that unless I didn't have any other options. Basically, I want to avoid becoming a full time cancer patient. As long as I still have some reasonable options, probably a clinical trial would be the last of my options that I would consider, unless it was more convenient, stage three, and relatively low risk.

My Treatment Principles

• Pursue approved drugs first
• Open to off-label protocols (e.g., Care Oncology)
• Look for combinatorial approaches
• Consider intermittent treatment or rechallenging treatments (e.g., ADT, chemo)
• Look for late-stage clinical trials

• Consider alternative treatments (e.g., diet [Filtricine])
• Refine supplement strategy

I would pursue approved drugs first. I'm open to off label protocols like the Care Oncology protocol. I've been on and off of that for a few years. I'm particularly interested in combinatorial or combination therapies or approaches that we haven't really had much conversation with my oncologist about, considering that it's pretty much been monotherapy. I would consider late stage clinical trials if they're not too cumbersome. I'm also open to alternative treatments. I am going to be a beta tester for the Filtracine diet, which is all prepared foods that exclude non-essential amino acids on the theory that cancer feeds on some non-essential amino acids. I think Brian tried that and found it wasn’t too palatable.

Brian McCloskey (25:02)

I did it for a week or two.

Robert Ellis (25:09)

I'm always game to try something like that. I have almost always been experimenting with various supplements. It's always difficult to know which ones are helpful or not in terms of disease, certainly, but I can usually tell whether or not they're helping with some of the side effects. I plan on being a little more intentional about diet and supplementation.

Testing Principles

• Identify targetable markers
• Whole genome
• RNA seq
• PSMA scan
• Future biopsy? (Original tissue unusable; recent biopsy negative)
• Bone biopsy?
• Liquid biopsy (Tempus xF+, 523 genes)
• Proteomics (mProbe; https://www.mprobe.com/oncoomicsdx)

With regard to testing, my main goal is to find some way of identifying more targetable markers that might give me more treatment options. I have not yet had a whole genome or RNA sequence. I will have a PSMA scan. It's possible that I would consider a future biopsy, although as we all know, that's not a very pleasant experience. I wasn't keen to rush right into trying again to get more tissue. But I will if I see that there's progression or something. The original

tissue was unusable, and the biopsy in October was negative. On getting a bone biopsy, it seems like there's a mixed sentiment about this. Dr. Koontz said that, in his experience, the bone biopsies are rarely usable. It's very difficult to get usable tissue in his opinion. It seems like that's more valuable if you can compare it with prostate tissue. So I don't know if it's really worth pursuing at this point. And then question mark here, I don't know what else I should be considering or what other testing options I may have.

Current Therapy

Rubraca
Orgovyx
Dutasteride
Metformin

The drugs that I’m on right now, other than supplements and diet and so forth, are Rubraca, which I seem to be responding well to. Because I'm starting the Filtricine diet tomorrow, I'm going to have labs today. So I'll get another PSA. I am on Orgovyx, and have been on all this time for ADT. I do take dutasteride, and that's for the frequency. It seems to help a little bit. I take Metformin.

Treatment Options

Therapy Option	Targets	Treatment Rationale	Risks & Unknowns
Lutetium -177	PSMA	• PSMA positive • Next PSMA scan 1/26/23 • Check SUV values (higher is better response)	
BAT	AR	• AR positive	• I have bone mets • Mixed results • Don't know if I have the right markers for a positive response • Consideration: Dr. Koontz is willing to try it

The next option that I'm talking about with Dr. Koontz is Lutetium. I know that I'm PSMA positive, but I'll have another scan which will give me a more accurate assessment of my current disease. I have talked to him about bipolar androgen therapy. I'm AR positive. I think it was the Caris report. It shows that I have bone mets. I understand that that may be a more painful journey, at least initially. There's some mixed results. I don't know if I have the right markers. I don't know if I have AR copy number gain. I know that with the research at Johns Hopkins, they

may be identifying some other markers that will predict whether or not I'm likely to respond. I don't know if I have that yet. But I am fortunate that Dr. Koontz is – after several conversations – now willing to try it with me if it comes to that, but it's likely to be a second choice after Leutetium since Leutetium seems to be pretty reliably effective, at least for some period of time. So those are the only two things that I've actually discussed with Dr. Koontz so far. I don't have any other real options in the wings. I'm going through a curation process with the supplements that I'm taking so I don't really have anything to report there now.

Amit Gattani (29:25)

How is your physical health? When you say a therapy failed, can you feel it besides your PSA marker?

Robert Ellis (29:45)

The way that I know that a treatment is failing is I start to get bone pain. If you look at this roller coaster ride here, whenever it gets above 20 or so, I start to have bone pain. I know when a treatment is failing because I get my PSA. Now I'm getting it every month or so. I was getting it every three weeks when I was getting chemo. So I knew immediately when my PSA was rising, but if I were not getting a PSA test, I know when my PSA gets in the 20s or so I start getting pain and it's failing. Overall, my health is pretty good. The chemo knocked out my physical energy. My muscles were just shot. I could barely walk. I have a steep driveway, and I could barely walk up the driveway to get my mail. Since I ended the chemo, I'm getting my physical energy back, and I'm able to exercise. I'm starting to get some of my functional strength back. But the Rubraca makes me sleepy. It might be a combination with the season, but I'm definitely sleeping about an hour a day more than I was before the rain and everything.

Amit Gattani (31:29)

If you go back to the PSA chart. Obviously when your PSA goes above 20, you can feel the bone pain. On your last change, where you were around two and a half PSA and decided that the chemo was failing, seems a little bit early to know if the chemo was actually failing. Because obviously these things are not always monotonic. At that point, before you switched from chemo, did you have physical symptoms to indicate that this thing is not working?

Robert Ellis (32:08)

It's kind of hard to see. But there are actually several PSAs where there's a definite trend. For example, there was an outlier bump, and then it came down again. There were five or six PSAs in a row that were creeping up. At that time, it was determined that it was better to try something else. One of the things that I am interested in is the idea of intermittent treatment. It's possible that I could, if I don't do chemo for a year or two, my hope was that I would get a year out of Rubraca. Dr. Koontz said that in his experience, his patients are getting more like six months. You know, so if I get six months to a year, if I get a year out of lutetium, it's possible, I might be

able to get a response again from chemo. I don't know. I'm intrigued by the idea of intermittent therapies. I'm intrigued by the idea of combination therapies. But I haven't really talked too much about that with Dr. Koontz. I haven't really known what to ask, other than generally.

Amit Gattani (33:29)

The reason I was asking is that I also work with UCSF, and one of the very early things that the UCSF team and oncologists told me is the analogy of the pony express. You have a finite number of ponies and you want to get the most mileage out of any pony that you can. Probably most of you have heard of that analogy. So don't change the treatment. It's absolutely necessary because it's harder to get back to that treatment. And you need a longer holiday, like you're saying, maybe a couple of years before the treatment may be responsive. I have lived through situations where my PSA is rising meaningfully for a period of time. I'm still on the therapy until it becomes obvious that physical symptoms or other or scans are also showing that the disease progression is meaningful, and we should go change the therapy. So just to extend the life out of any therapy that we are on because there are only a finite number of therapy options that we have. That's where I was going with it. This looked like maybe chemo wasn't working as effectively as it could be. But has it failed, and is it necessary to switch? This was the decision. And to your point, I have a docetaxel rechallenge in my journey. I started my journey with docetaxel. And then after two years of other things, we came back when we needed to do chemo. We did docetaxel again. And it gave me about six or eight additional months. It gave me a moderate response. So the challenge with some of those things after a period of a holiday can be useful. It's been my personal experience.

Brian McCloskey (35:54)

I have a few thoughts regarding your current treatment options. On Lutetium: do you have high SUV values, which you should probably have picked up from your PSMA PET?

Robert Ellis (36:11)

I don't know. I'll see if I can find that somewhere.

Brian McCloskey (36:15)

That'll give you some sort of indication as to whether or not you would be a good candidate for PSMA Lu.

Robert Ellis (36:25)

The higher the SUV, the more likely I am to respond?

Brian McCloskey (36:32)

Correct. Standard Uptake Value. There should be a few different readings across the imaging field. You'll get variation. That's something to look at in terms of whether or not you're going to respond to that, because obviously you don't want to take a drug that isn’t going to help you.

With respect to the AR pathway, I'm wondering whether or not you would respond to an AR degrader. You have a high expression of AR, which is noted in your Caris report. And so I wonder if that would be something to consider as well. And as far as I know, there's ARV-110, and ARV-766 (a Phase 1/2 clinical trial to evaluate the safety and efficacy of ARV-766 given by mouth in men with metastatic castration-resistant prostate cancer who have progressed on prior approved systemic therapies). I'm not sure which one is the latest and greatest. You'd think that 766 is the higher number, but I'm not exactly sure.

There might be a little bit of a cheat sheet for you – my 20 different treatment options. Some of the options that apply to me might be worth talking to your physicians about as well. It's a bunch of stuff.

I'm going to look into this state of the art of bone biopsies and see if we can try to find some new information on that. It's just so fundamental to what we're trying to do.

I know that Peter Kuhn has talked about his liquid biopsy that requires a bone aspirate. I don't know if that technology is there. I'll do a little bit of digging for you and see what I can come up with on the whole bone biopsy business.

Robert Ellis (38:59)

I'd love to get an updated CureMatch report. I don't know if I should wait until after my PSMA.

Ally Perlina (39:21)

Happy to help. There have been a lot of updates in general in the field and in our reports, so maybe even with the same inputs, it would look slightly different if we produced it now. So maybe there's still a point to run it. But if there's new incoming data, then for sure, it's even more important to run it.

Brad Power (40:46)

CureMatch and Ally work off of biomarkers, which you have. The PSMA test is more like a scan. It's not going to give you more biomarkers, so it's not going to add anything beyond your PSMA status, because you had liquid biopsy results. If you had other biomarker indicators, that's where they start as their basis.

Ally Perlina (41:11)

We take PSMA as one of the markers, but it's more like we either consider it as part of the inputs or not, it's not sensitive to the levels of PSMA. In that sense, it won't make a difference if it's just a little bit higher, or a little bit lower. But it would trigger certain matching, which maybe in 2021 wasn't there.

Robert Ellis (41:32)

Are any of these other biomarkers targetable? I have no idea what any of these are, such as MSS. I'm looking at my chart: MSS, PD-L1, TMB. I suppose if you run a report you would be able to tell me if those are targets?

Ally Perlina (41:59)

Yeah, but I can already tell you that there would be some potential combination therapies just looking at the markers. They would trigger some drugs for you.

Robert Ellis (42:08)

Okay, perfect. Thank you so much.

Rick Stanton (42:11)

Everyone here should know what those mean eventually. For example, MSS is microsatellite stability. That's being stable. High microsatellite instability is an indication of a lot of mutations.

Robert Ellis (43:01)

I will do some research.

Does anyone have any other thoughts about what I should be looking for?

Ally Perlina (43:14)

How old is that PDL result? When was it determined that there was some low expression?

Robert Ellis (43:23)

This is from the original tissue. So that would have been October 2017. So it's pretty old.

Ally Perlina (43:36)

Is there any way to get your PDL1/PD1 assessment and maybe the current tumor mutation burden as well as MSS/MSI assessment.

Robert Ellis (43:49)

Can that be obtained from a liquid biopsy?

Ally Perlina (43:54)

They do some of that at least nowadays. Yes. I've seen that in reports from Tempus, maybe Foundation as well. So TMB for sure.

Robert Ellis (44:11)

If I have no tissue, and I can only do a liquid biopsy, what exactly should I request from my oncologist?

Ally Perlina (44:22)

Let me look into that. That's a good question, because they've come up with some new ways to evaluate the same mark.

Robert Ellis (44:30)

I know that Pacific Cancer Care now works with Tempus. What exactly is the test that I should request?

Rick Stanton (44:55)

Tempus xE is the whole exome test (19,000 genes). xF is the liquid biopsy on 105 mutational hotspots.

Brian McCloskey (45:26)

Tempus has a new xF+ test, which is a 523-gene panel liquid biopsy (blood test).

Robert Ellis (45:41)

I should be able to get that.

Brian McCloskey (45:43)

I'm going to reach out to Tempus and FMI to find out about the bone stuff, but definitely you should be able to get the liquid biopsy. Tempus has the xF+. That's also in my data profile. If you look at my data docs, you'll see that.

Brad Power (46:05)

Mike Yancey just had an mProbe proteomics report this past week. That's another test that Robert may want to consider.

Mike Yancey (46:16)

That one doesn't take a lot of tissue. They were able to pull tissue from the lesion on my spine. We first sent tissue to Tempus. We wanted to make sure that Tempus got everything they needed to do the whole exome sequencing and the RNA sequencing. Then we sent it to mProbe. Once they got it, it took them less than a week to do their work and get back with me. They were very easy to work with. They gave me a great report. I've got an expression of one protein that Sheeno says is the highest he's ever seen in prostate cancer. They offered some recommendations for potential drugs to use. so that's a conversation the oncologist and I still need to have. mProbe is a great company to work with based on what I've seen so far. And the ease of interface.

Robert Ellis (47:25)

I’ll go there. They need tissue for that? Did you have cancer in your spine?

Mike Yancey (47:34)

Yes. We had a growth attached to my spinal cord, so we did a biopsy on that.

Robert Ellis (47:39)

I don't know where they would get tissue for me. I'd have to do another prostate biopsy or bone biopsy.

Mike Yancey (47:49)

You can see the mProbe presentation by Sheeno on the website from several months ago.

Robert Ellis (47:55)

I'll definitely look into that. Thank you.

Brad Power (48:02)

On the comment about staying with a therapy that's working: Rick, maybe you could report on your experience of working with Dr. Gatenby's team through Dr. Dawn Lemanne. It seems they're almost on the other end of the spectrum, which is if your PSA is not high, if you have “altitude” as they called it, and you have a number of therapy options, run an experiment with each of them, just to see which one is working. Could you speak to that as an alternative?

Because what Amit was pointing at there is a strategy that is overarching to the treatments you're trying,

Rick Stanton (48:44)

I also stopped docetaxel on my 11th round as my PSA was creeping up. That decision wasn't because my disease was wildly progressing. It was because I was beat up. I could barely do one push up. Just like you barely made it up your driveway. So it was a decision to take an easy-to-get-to drug, and let's see if we can get some mileage elsewhere because I was in trouble. Not so much from the disease, but from my quality of life being diminished.

Dr. Robert Gatenby has the opinion of avoiding maximum tolerable dose until failure. He's trying to look through an evolutionary lens. I was deemed castrate resistant after failing casodex and daralutamide. I went on docetaxel for almost a year. I was running out of options, and I was waiting for Pluvicto. That's when (“Dr.”) Bryce Olson said, “Why don't you try abiraterone? You’ve got nothing to lose, and it’s easy to take.” My doubling time was about a month. Abiraterone worked. So am I now castrate sensitive? We all know that this cancer moves. According to Dr. Gatenby the sensitive population during the time of lack of pressure, during chemo, rebounded, and now I'm sensitive again. I'm throwing that out to you. Just because you were deemed castrate resistant doesn't mean that in the evolutionary theory that it will stay that way, if you stop pushing on it.

Robert Ellis (51:16)

I'm very intrigued by Gatenby’s ideas. Were you on any ADT at all? I've been on Orgovyx (an androgen deprivation drug) all this time. Maybe I should stop Orgovyx

Rick Stanton (51:31)

I was off of ADT. I went to a PDL1 and adenosine inhibitor, and it didn't work at all. That's when I went back to ADT. I'm seeing Dr. Lemanne, an integrative oncologist out of Oregon, who works with Dr. Gatenby. Dr. Gatenby, wants to have an MD as the prescriber. I find her inputs very complementary to Dr. McKay. I recommend anyone who's interested in that, or pulsing ADT, to reach out to me or look up Dr. Lemanne. https://www.oregonio.com/
Dr. Lemanne did a number of tests, including bloodwork and other tests, that were well outside UCLA or UC San Diego, and that helped assess my natural killer cell function. Not only the quantity, but their function. It costs me a couple $1,000 to get access to her time.

Robert Ellis (53:13)

This is telemedicine. I'll check that out. Thanks.

Brian McCloskey (53:38)

On the question about SUV values: What is considered a high SUV value? I know that Rick and Mike are both on Lutetium. I recall that it was a number over 10. I know Mike, yours were off the chart. You were like a 20 or something crazy. Can you guys add some thoughts on that?

Mike Yancey (54:08)

My lowest was some spots that were putting out closer to 10, which is adequate. But I also had a lot of spots around 20, 30, or 40. That was very, very significant. Looking back – I'm getting ready to do my fifth infusion – it seems to be working quite well. Based on my SUV values, it seems to correlate pretty well.

Brian McCloskey (54:37)

Rick, do you recall your SUV values?

Rick Stanton (54:41)

My SUV max was 11.

Brian McCloskey (54:44)

You talked about clinical trials as something you want to go after, but you don't want to do stage one trials. I would recommend that you try Massive Bio. It's pretty quick. You can take all of your information that's on the Google Drive, upload it, and then they'll turn around and report pretty quickly. You could probably put in parameters like you’re not interested in travel. That's definitely a parameter. You may be able to specify what stage, and they'll work with you. They're awesome. Their suggestions definitely made it down to the top three for me.

Robert Ellis (55:36)

Thanks for that. I probably made a mistake by waiting for the PSMA scan, and they did the wrong scan, then waiting for the biopsy, and then they didn't get any tissue. So I've been in a little bit of limbo since fall. I'll do that.

Brian McCloskey (56:00)

Here is the link that's specific to us at Massive Bio (PCL-only registration page):
https://share.hsforms.com/1u_n-eW_jQtCeK0DvZGO6fQ1leeg
Click on this URL. It is handy. I just sent it off to Amit yesterday. This is specific for Prostate Cancer Lab patients. So try that door.

Robert Ellis (56:21)

I can't seem to find my PSMA results. I'm not sure where they are. I'll see if I can find that and find out what it says my SUV is. If not, I'll know in a few in a month or so.

Brian McCloskey (56:56)

I just started Provenge yesterday, the leukapheresis process. You might want to talk to your doctor about that.

Robert Ellis (57:11)

I had Provenge. Sorry, I don't know how I missed that. I know all about squeezing the ball for three hours.

I had to go to San Jose for the leukapheresis, and then drive up to San Francisco for the infusions. We binge watched Jason Bourne movies while we were sitting.

Brian McCloskey (57:44)

Do you remember what your PSA was? Because supposedly it works better if your PSA is low.

Robert Ellis (57:53)

I don't. It was early in 2021. It was after I became castrate resistant. It was probably under 10, around six, somewhere in there.

The problem with Provenge is there's no real way to know whether or not it's working, it doesn't affect your PSA. It won't bring your PSA down. It’s just based on the research that you get some mileage from it.

Brian McCloskey (58:35)

I'm trying to get Saul Priceman to join us. He's at City of Hope and an expert in personalized vaccines, particularly CAR-T.

Robert Ellis (59:04)

I am so grateful for all your input. Thank you so much.

—

Robert’s CureMatch Report

“Testing and Treatment Options Review for Robert Ellis” (Robert Ellis) [#39]

Ally Perlina ran a fresh CureMatch report for Robert on January 14, 2023, which showed their usual three-drug, two-drug, and single drug matches with Robert’s four actionable biomarkers: APC, BRCA2, AR, and FOLH1 (PSMA).

The three three-drug combinations were:

- abiraterone + olaparib + sulindac (49%)
- carboplatin + degarelix + sulindac (49%)
- apalutamide + lutetium lu 177 vipivotide tetraxetan + sulindac (39%)

The two-drug combinations were:

- abiraterone + olaparib (39%)
- carboplatin + degarelix (39%)
- apalutamide + lutetium lu 177 vipivotide tetraxetan (29%)

The single drugs were:

- Abiraterone (20%)
- Olaparib (20%)
- lutetium lu 177 vipivotide tetraxetan (10%)

Unfortunately, Robert has already been treated with most of these, either as mono or two-drug combinations, but he will discuss with his doctor if he can rechallenge with them.